Final Exam

Graded

Student

Tieu Nghi

Total Points

77 / 100 pts

Question 1

Question 1 **14** / 18 pts

1.1 a **10** / 10 pts

✔ **− 0 pts** Correct

− 2 pts first is wrong

− 2 pts second is wrong

− 2 pts third is wrong

− 2 pts fourth is wrong

− 2 pts fifth is wrong

1.2 b **4** / 8 pts

− 0 pts Correct

− 4 pts sigma is positive

✔ **− 4 pts** Q is negative

− 8 pts not done or wrong

− 3 pts explanation is wrong

Question 2

Question 2 — **16** / 16 pts

2.1 a — **8** / 8 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** COP is wrong
- **– 2 pts** Kelvin scale is not used
- **– 8 pts** not done or wrong
- **– 3 pts** COP equation is wrong

2.2 b — **8** / 8 pts

- ✔ **– 0 pts** Correct
- **– 3 pts** h2s=2833
- **– 2 pts** Equation is wrong
- **– 8 pts** not done or wrong
- **– 2 pts** h1=3192
- **– 2 pts** h1-h2=100

Question 3

Question 3 **31** / 32 pts

3.1 a **2** / 2 pts

✔ **− 0 pts** Correct

3.2 b **3** / 3 pts

✔ **− 0 pts** Correct

− 1 pt closed system

3.3 c **4** / 4 pts

✔ **− 0 pts** Correct

− 2 pts energy balance is wrong

− 2 pts entropy balance is wrong

3.4 d **8** / 8 pts

✔ **− 0 pts** Correct

− 1 pt v2=0.4015

− 1 pt P=2.7 bar

− 1 pt unit conversion is wrong

− 2 pts need to multiply with mass

− 1 pt u2 is 1742

− 1 pt v1= 1.069x10^-3

− 1 pt u1= 546.02

− 4 pts Q is not calculated or equated to W. (Q is not equal to work in this example)

− 8 pts not done or wrong

− 3 pts work equation is wrong

3.5 e **5** / 5 pts

✔ **− 0 pts** Correct

− 1 pt s2=4.86

− 1 pt s1=1.634

− 2 pts need to multiply with mass

− 5 pts not done or wrong

3.6 **f** **5** / 5 pts

- ✔ **– 0 pts** Correct
- **– 1 pt** T= (20+273)K
- **– 1 pt** calculation error
- **– 3 pts** equation is wrong
- **– 5 pts** not done or wrong

3.7 **g** **4** / 5 pts

- **– 0 pts** Correct
- **– 5 pts** not done or wrong
- **– 1 pt** s2 on TS is wrong
- **– 2 pts** s1 in both are wrong
- **– 1 pt** s1 in Pv wrong
- ✔ **– 1 pt** s2 on Pv wrong

Question 4

Question 4 **16** / 34 pts

4.1 a **4** / 4 pts

✔ **− 0 pts** Correct

− 1 pt steady state

− 1 pt control volume

− 4 pts not done or wrong

4.2 b **6** / 6 pts

✔ **− 0 pts** Correct

− 6 pts not done

4.3 c **6** / 6 pts

✔ **− 0 pts** Correct

− 6 pts not done or wrong

− 3 pts equation is wrong

− 1 pt calculation error

4.4 d **0** / 6 pts

− 0 pts Correct

− 3 pts can not use Temperatures

✔ **− 6 pts** not done or wrong

− 1 pt calculation error

4.5 e **0** / 6 pts

− 0 pts Correct

− 1 pt calculation error

− 3 pts equation is wrong

✔ **− 6 pts** not done or wrong

4.6 f

0 / 6 pts

− 0 pts Correct

− 1 pt calculation error

✔ − 6 pts not done or wrong

− 3 pts s1 s2 not calculated

CHE 201 Final Exam **NAME:** Tieu Nghi

Bilgin December 9^{th}, 2025

Total points: 100

Please fill in your answers in the space supplied

Show all of your calculations, conversions and work to receive full credit.

QUESTION 1 (18 pts)

True or False (2 pts each)

False Energy transferred by heat to a closed system during an irreversible system can be calculated by the area under the curve in T-v diagram.

True Entropy, **S**, is ~~extensive~~ intensive property. $S = \frac{Q}{T}$

False An ideal gas in a piston-cylinder assembly expands isothermally, doing work and receiving an equivalent amount of energy by heat transfer from the surrounding atmosphere. This process of the gas is in violation of the Kelvin-Planck statement of the second law.

False The maximum thermal efficiency of *any* power cycle operating between hot and cold reservoirs at 1000°C and 500°C, respectively, is 50%.

True The change in entropy of a closed system is the same for every process between two specified end states.

Fill in the blank

(8 pts) A closed system undergoes an internally irreversible process in which work is done on the system and the heat transfer occurs from the system only at a location on the boundary where the temperature is T_b. Determine whether the entropy change of the system is positive, negative, zero, or indeterminate. Explain.

Irreversible process $\rightarrow \sigma > 0$, $S \propto \frac{1}{T}$.

Closed system $\rightarrow \Delta S = S_2 - S_1 > 0$.

$\Rightarrow S_2 > S_1$.

$\Delta S = \int \frac{\delta Q}{T} + \sigma$

$\boxed{\therefore \text{The entropy chang of the system is positive.}}$

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QUESTION 2 (16 pts)

a) (8 pts) An inventor has developed a refrigerator capable of maintaining its freezer compartment at 20°F while operating in a kitchen at 70°F, and claims the device has a coefficient of performance of 10. Evaluate the claim.

T_c T_H

$\beta_{max} = \frac{T_c}{T_H - T_c} = \frac{479.67\ °R}{(529.67 - 479.67)°R} = 9.5934$

- $20°F = 266.5\ °K = 479.67\ °R$
- $70°F = 294.26\ °K = 529.67\ °R$

$\beta = \frac{Q_c}{W_e} = 10$

$$\boxed{(\beta = 10) \approx (\beta_{max} = 9.5934) \simeq 9.6 \simeq 10}$$

∴ Inventor has developed a fridge capable of maintaining it freezer compartment at 20°F while operating in a kitchen at 70°C, the device has a coefficient of performance of 10.

b) (8 pts) Water vapor enters a turbine at 3 bar $T_1 = 360°C$ and exits at $p_2 = 0.7$ bar. The work developed is measured as 100 kJ per kg of water vapor flowing. Calculate the isentropic turbine efficiency.

$\eta = \frac{(h_1 - h_2)}{h_1 - h_{2s}}$ = 100 kJ/kg

$P_1 = 3$ bar, $T_1 = 360°C$ $\xrightarrow{\text{Superheat water table}}$ $h_1 = 3192.2$ kJ/kg, $s_1 = 7.9061$ kJ/kg.K

$W = h_1 - h_2 = 100$ kJ/kg

$P_2 = 0.7$ bar $\xrightarrow{\text{table A-3}}$ • h_2: $h_{2g(0.7bar)} = 2660.0$ kJ/kg

$h_{2f(0.7bar)} = 376.70$ kJ/kg

• $s_{2s} = s_1 = 7.9061$ kJ/kg.K

$s_{f(0.7bar)} = 1.1919$ kJ/kg.K

$s_{g(0.7bar)} = 7.4797$ kJ/kg.K

$\Rightarrow x_2 = \frac{s_{2s} - s_f}{s_g - s_f} = \frac{7.9061 - 1.1919}{7.4797 - 1.1919}$

$\Rightarrow x_2 = 1.0678$

• $h_{2s} = h_{2f} + x_2(h_{2g} - h_{2f})$

$= 376.7 + 1.0678(2660 - 376.7)$

$\Rightarrow h_{2s} = 2814.8$ kJ/kg

$\Rightarrow \eta = \frac{100\ kJ/kg}{(3192.2 - 2814.8)\ kJ/kg}$

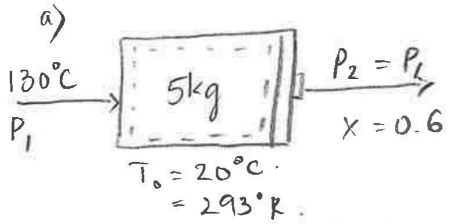
$\Rightarrow \boxed{\eta \simeq 0.265 = 26.5\%}$

QUESTION 3: (32 pts) $\nearrow P_{sat} = 2.701$ bar.

5 kg of water, initially a saturated liquid at 130°C is contained piston-cylinder device. The water undergoes a process. Heat is transferred between the piston-cylinder and a 20°C room until the water becomes a state where x=0.6. The piston moves freely and maintains constant pressure throughout the process. Kinetic and potential energy can be ignored.

~~a)~~ (2 pts) Draw a schematic of the system and label the system boundary.
~~b)~~ (3 pts) State the engineering model
~~c)~~ (4 pts) Write the simplified energy balance and simplified entropy balance.
~~d)~~ (8 pts) Determine the work in **kJ**, and the heat transferred in **kJ**.
~~e)~~ (5 pts) Determine the change in entropy of the water during the process, **ΔS, in kJ/K**
f) (5 pts) Determine the entropy generated during the process, **σ, in kJ/K.** State if this process is reversible, irreversible or impossible.
g) (5 pts) Carefully sketch the process on P-v and T-s diagram.

a)



b) Model:

piston - cylinder $\rightarrow$ Closed system.
state1 : sat. liquid.
state2 : mix
water undergoes constant pressure
KE = PE = 0

c). Closed system:

$\Delta E = Q - W$

$\Leftrightarrow \cancel{\Delta KE} + \cancel{\Delta PE} + \Delta U = Q - W$.

$\Rightarrow \boxed{\Delta U = Q - W}$. (energy balance)

• Entropy balance:

$\Delta S = \int_1^2 \frac{\delta Q}{T} + \sigma$

$\Leftrightarrow \boxed{S_2 - S_1 = \frac{Q}{T_0} + \sigma}$.

@ sat. liq, $T = 130°C$, $P_{sat} = 2.701$ bar.

$h_1 = h_{f(130°C)} = 546.31$ kJ/kg.

$u_1 = u_f(130°C) = 546.02$ kJ/kg.

$s_1 = s_f(130°C) = 1.6344$ kJ/kg.K.

$v_1 = 1.0697 \times 10^{-3}$ m³/kg

@ state 2, x = 0.6

$s_2 = s_{2f} + x(s_{2g} - s_{2f})$

$= (1.6344) + (0.6)(7.0269 - 1.6344)$.

$\Rightarrow s_2 = 4.8699$ kJ/kg.K.

$v_2 = v_{1f} + x(v_{1g} - v_{1f}) \Rightarrow v_2 \doteq 0.4015$ m³/kg

$= 1.0697\times10^{-3} + (0.6)(0.6685 - 1.0697\times10^{-3})$

$u_2 = u_{2f} + x(u_{2g} - u_{2f})$

$= 546.02 + (0.6)(2539.9 - 546.02)$

$\Rightarrow u_2 = 1742.348$ kJ/kg.

d). $W = \int_1^2 P\,dv = Pm(v_2 - v_1) = (2.701\text{ bar})(5\text{ kg})(0.4015 - 1.0697\times10^{-3})\ m^3/kg.$

$\Rightarrow \boxed{W = 5.408\ \text{bar}\cdot m^3}$
$\Rightarrow \boxed{W = 540.8\ kJ}$

$\Delta U = Q - W$

$\Rightarrow Q = \Delta U + W = m(u_2 - u_1) + W.$

$= 5kg(1742.35 - 546.02)\frac{kJ}{kg} + 540.8\ kJ.$

$\Rightarrow \boxed{Q = 6522.45\ kJ}$

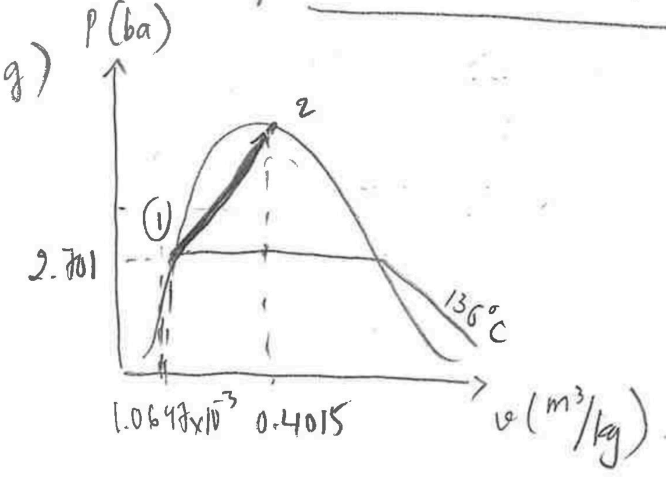
e) $\Delta S = m(s_2 - s_1) = 5kg(4.8699 - 1.6344)\ kJ/kg.K$

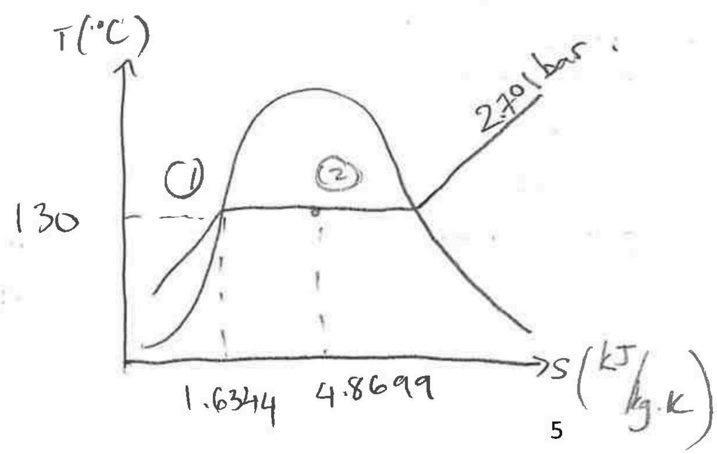
$\Rightarrow \boxed{\Delta S = 16.178\ kJ/K}.$

f) $\sigma = \Delta S - \frac{Q}{T_0} = 16.178\frac{kJ}{K} - \frac{-6522.45\ kJ}{(20^\circ C + 273)^\circ K}.$

$\Rightarrow$ $\sigma = 38.439\ kJ/K$
$\sigma > 0 \Rightarrow$ Irreversible processe
$Q < 0$ due to compress.

g)





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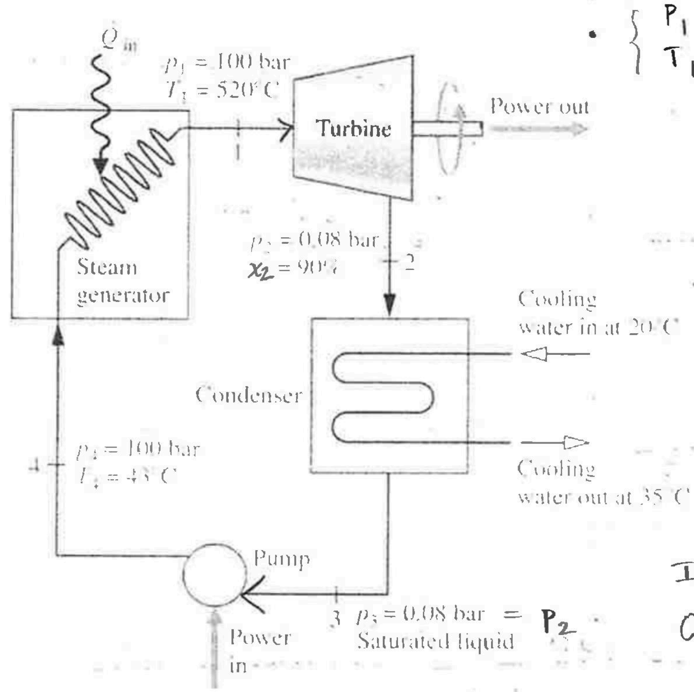
QUESTION 4: (34 pts)

The figure below shows a simple vapor power plant operating at steady state with water as the working fluid. Data at key locations are given on the figure. The mass flow rate of the water circulating through the components is 109 kg/s. Stray heat transfer and kinetic and potential energy effects can be ignored.



• $\begin{cases} P_1 = 100\,bar \\ T_1 = 520°C \end{cases} \xrightarrow{\text{superheated}}$ water (A-4)

$v_1 = 0.03394\ m^3/kg$
$u_1 = 3085.6\ kJ/kg$
$h_1 = 3425.1\ kJ/kg$
$s_1 = 6.6622\ kJ/kg.K$

	Turbine	Pump
Inlet: $\dot{m}_i$ =	$\dot{m}_1$ &	$\dot{m}_3$
Outlet: $\dot{m}_e$ =	$\dot{m}_2$ &	$\dot{m}_4$

(a) (4pts) State the engineering model.
(b) (6pts) Write simplified balances for mass, energy and entropy.
(c) (6pts) Calculate the net power developed, in MW.
(d) (6pts) Calculate the thermal efficiency.
(e) (6pts) Determine the mass flow rate of the cooling water, in kg/s.
(f) (6pts) Determine the rates of entropy production, in kW/K, for the turbine.

a) Vapor power plant
Steady state
$\dot{Q} = 0$
$KE = PE = 0$
Open system

b). Mass balance $\cancel{\frac{dm_{cv}}{dt}} = \sum \dot{m}_i - \sum \dot{m}_e$
S.S: 0

$0 = \sum \dot{m}_i - \sum \dot{m}_e$

$\Rightarrow \boxed{\dot{m}_i = \dot{m}_e}$

• Energy balance: $\cancel{\frac{dE_{cv}}{dt}} = \dot{Q} - \dot{W} + \sum \dot{m}_i(h_i + \cancel{\frac{V_i^2}{2}} + \cancel{gz_i}) - \sum \dot{m}_e \times (h_e + \cancel{\frac{V_e^2}{2}} + \cancel{gz_e})$

$\Leftrightarrow 0 = -\dot{W} + \dot{m}_i h_i - \dot{m}_e h_e$

$\Leftrightarrow \boxed{\dot{W} = \dot{m}_i h_i - \dot{m}_e h_e} \Rightarrow \boxed{\dot{W} = \dot{m}(h_i - h_e)}$

b) (cont.).

Entropy balance:

$$\cancel{\frac{dS_{cv}}{dt}} = \cancel{\sum \frac{\dot{Q}}{T_0}} + \sum \dot{m}_i s_i - \sum \dot{m}_e s_e + \dot{\sigma}_{cv}$$

$$(\Rightarrow) \boxed{0 = \dot{m}_i s_i - \dot{m}_e s_e + \dot{\sigma}_{cv}}$$

$$h_4 = h_f(43°C): \frac{43-40}{h_4 - 167.57} = \frac{45-40}{188.45 - 167.57}$$

$$\Rightarrow h_4 = 180.098 \text{ kJ/kg}.$$

c) • $P_3 = 0.08$ bar, sat. liquid:

Table A-3: $v_3 = v_{3f(0.08)} = 1.0084 \times 10^{-3}$ m³/kg

$$u_3 = u_{3f(0.08)} = 173.87 \text{ kJ/kg}.$$

$$s_3 = s_{3f(0.08)} = 0.5926 \text{ kJ/kg.K}$$

$$h_3 = h_{3f(0.08)} = 173.88 \text{ kJ/kg}.$$

• $P_2 = 0.08$ bar, $x = 90\% = 0.9$

$T_{sat} = 41.51 \rightarrow$ Mix

$$h_2 = h_{2f} + x(h_{2g} - h_{2f})$$
$$= 173.88 + (0.9)(2577.0 - 173.88)$$
$$\Rightarrow \boxed{h_2 = 2336.688 \text{ kJ/kg}.}$$

$$\left\{\begin{array}{l} v_{2f} = 1.0084 \times 10^{-3} \text{ m}^3/\text{kg} \\ v_{2g} = 18.103 \text{ m}^3/\text{kg} \end{array}\right. \longrightarrow v_2 = v_{2f} + x(v_{2g} - v_{2f})$$
$$= 1.0084 \times 10^{-3} + (0.9)(18.103 - 1.0084 \times 10^{-3})$$
$$\Rightarrow \boxed{v_2 = 16.293 \text{ m}^3/\text{kg}}.$$

$$\left\{\begin{array}{l} u_{2f} \quad 173.87 \text{ kJ/kg}. \\ u_{2g} \quad 2432.2 \text{ kJ/kg}. \end{array}\right. \longrightarrow u_2 = u_{2f} + x(u_{2g} - u_{2f})$$
$$= 173.87 + (0.9)(2432.2 - 173.87)$$
$$\Rightarrow \boxed{u_2 = 2206.367 \text{ kJ/kg}}$$

$$\left\{\begin{array}{l} s_{2f} \quad 0.5926 \text{ kJ/kg.K} \\ s_{2g} \quad 8.2287 \text{ kJ/kg.K} \end{array}\right. \longrightarrow s_2 = s_{2f} + x(s_{2g} - s_{2f})$$
$$= 0.5926 + (0.9)(8.2287 - 0.5926)$$
$$\Rightarrow \boxed{s_2 = 7.465 \text{ kJ/kg.K}}.$$

• $P_4 = 100$ bar $\rightarrow T_{sat} = 311.1$ °C

$T_4 = 43$ °C $< T_{sat}$

Compressed liquid: (Table A-2)

$$\cdot\ v_4 = v_f(43°C): \frac{43-40}{v_4 - 1.0078 \times 10^{-3}} = \frac{45-40}{(1.0099 - 1.0078) \times 10^{-3}} \Rightarrow v_4 = 0.00100906 \text{ m}^3/\text{kg}.$$

$$\cdot\ u_4 = u_f(43°C): \frac{43-40}{u_4 - 167.56} = \frac{45-40}{188.44 - 167.56} \Rightarrow u_4 = 180.088 \text{ kJ/kg}.$$

$$\cdot\ s_4 = s_f(43°C): \frac{43-40}{s_4 - 0.5725} = \frac{45-40}{0.6387 - 0.5725} \Rightarrow s_4 = 0.6122 \text{ kJ/kg.K}.$$

$$\dot{W} = \dot{m}(h_i - h_e)$$

$$= \dot{m}[(h_1 + h_3) - (h_2 + h_4)]$$

$$= 109\ \frac{kg}{s}\left[(3425.1 + 173.88) - (2836.688 + 180.098)\right] kJ/kg$$

$$\Rightarrow \boxed{\begin{aligned} \dot{W} &= 117959.146\ kJ/s \\ &= 117959.146\ kW \\ &= 117.96\ MW \end{aligned}}$$

d)